VTTSI-Chat: Real-Time Intersection Safety Monitoring with LLM-Powered Natural Language Explanation

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***Abstract*—We present VTTSI-Chat, a cloud-native system that pairs the Virginia Tech Transportation Safety Index (VTTSI)—a hybrid Real-Time Safety Index (RT–SI) and CRITIC-weighted MCDM pipeline—with a SafetyChat conversational module powered by a tool-augmented GPT-4o. VTTSI fuses seven years of VDOT crash records via Empirical Bayes stabilization with live connected-vehicle telemetry, producing interpretable 0–100 safety scores every 15 minutes. SafetyChat exposes six typed API tools to the LLM, grounding all natural language responses in live data and enabling TMC operators, planners, and emergency responders to query causal safety narratives through ordinary conversation. A live deployment at https://cs6604-trafficsafety-180117512369.europe-west1.run.app demonstrates four end-to-end use cases across monitored intersections in Falls Church, VA.**

I. Introduction

TMC operators must interpret multi-dimensional safety indices under time pressure, yet existing dashboards present data without explaining causality or suggesting actions. Tool-augmented LLMs can reason over structured numerical data to produce grounded explanations [9], but prior LLM+traffic work targets signal control [4] or AV motion planning [5, 12]—not safety-index explanation for infrastructure operators.

VTTSI-Chat closes this gap by pairing the VTTSI hybrid scoring pipeline with **SafetyChat**: a GPT-4o module that issues typed function calls to live VTTSI endpoints and returns causal safety narratives via natural language dialogue. It is the first system to combine a hybrid EB-stabilized crash index with real-time MCDM scoring *and* an operator-facing LLM explanation layer.

II. VTTSI System Overview

The VTTSI backend (FastAPI, Google Cloud Run) computes two complementary safety indices every 15 minutes from connected-vehicle telemetry streamed via the VTTI Trino system and stored in Cloud SQL PostgreSQL.

A. Real-Time Safety Index (RT–SI)

RT–SI anchors intersection risk to a severity-weighted Empirical Bayes (EB) baseline derived from 2017–2024 VDOT crash records. The historical crash rate is:

$$r_i = \frac{\sum_s w_s\, C_{i,s}}{E_i}, \quad w_{\text{Fatal}} = 10,\ w_{\text{Inj}} = 3,\ w_{\text{PDO}} = 1, \quad (1)$$

where $C_{i,s}$ are crash counts by severity and E_i is vehicle exposure. The EB stabilized rate is:

$$\hat{r}^{\text{EB}}_{i,t}(\lambda) = \frac{Y_{i,t} + \lambda\, r_0}{1+\lambda}, \quad \lambda^\star = 10{,}000, \quad (2)$$

where r_0 is the global mean rate across all site-bins (2017–2024) and $\lambda^\star$ is chosen by minimizing Poisson log-loss on a 2025 hold-out set. Real-time uplift factors—bounded to $[0, 1]$—adjust this baseline for speed deficit (F^{spd}), speed variance (F^{var}), and VRU–vehicle conflict exposure (F^{conf}):

$$U_{i,t} = 1 + \beta_1 F^{\text{spd}}_{i,t} + \beta_2 F^{\text{var}}_{i,t} + \beta_3 F^{\text{conf}}_{i,t}. \quad (3)$$

VRU and vehicle sub-indices are multiplied by $U_{i,t}$ and the EB baseline, combined with weights ω_{vru} and ω_{veh}, and min-max scaled to produce $\text{SI}^{\text{RT}} \in [0, 100]$.

B. MCDM Safety Index

A CRITIC-weighted multi-criteria pipeline [1] constructs a decision matrix over five real-time criteria (vehicle count, VRU count, average speed, speed variance, incident count) for each intersection–time-bin pair. CRITIC criterion weights are derived from normalized standard deviations and inter-criterion Pearson correlations, recomputed every 24 hours. Three MCDM scoring methods—SAW, EDAS, and CODAS—are each computed and scaled to $[0, 100]$; a second-pass CRITIC aggregation yields the hybrid index $\text{SI}^{\text{MCDM}} \in [0, 100]$ (higher = higher risk).

C. Blended Final Index

$$\text{SI}^{\text{Final}}_{i,t} = \alpha\, \text{SI}^{\text{RT}}_{i,t} + (1-\alpha)\, \text{SI}^{\text{MCDM}}_{i,t}, \quad \alpha = 0.7 \text{ (default)}. \quad (4)$$

Scores map to risk levels: 0–40 = low, 41–70 = moderate, 71–100 = high. The α parameter is user-adjustable to shift emphasis between historical risk and instantaneous anomaly detection.

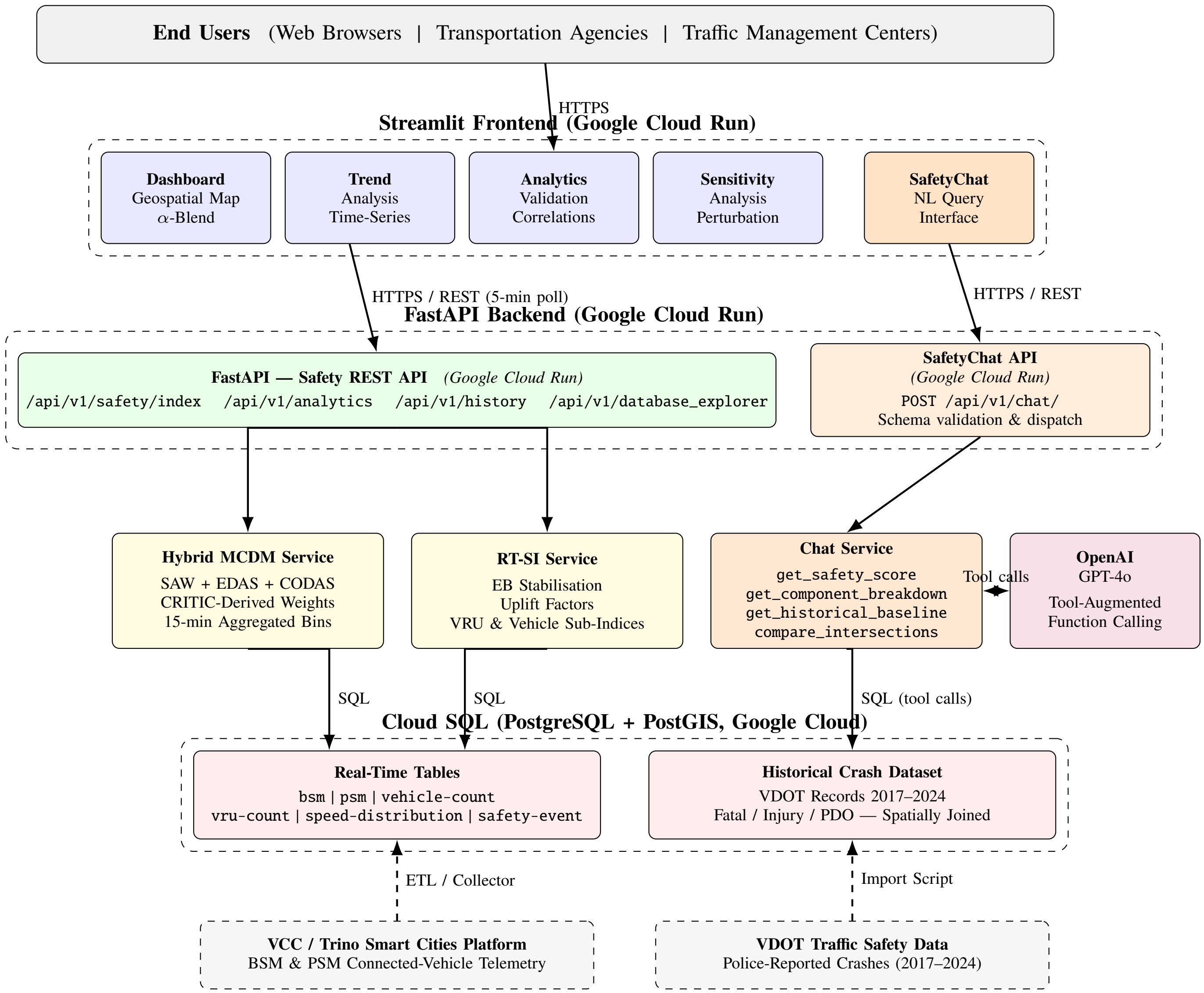

Fig. 1. VTTSI cloud-native system architecture. The orange pathway (SafetyChat API, Chat Service, OpenAI GPT-4o) is the new contribution of this demo paper; the blue/green/yellow left pathway shows the existing safety-index pipeline.

D. Geospatial Data Foundation

Each intersection is registered with a centroid (lat_i, lon_i) and an axis-aligned bounding box B_i ($\delta \approx 0.001°$, ≈ 111 m). VDOT crash records are assigned to intersections at import time via a PostGIS `ST_Within` predicate; overlaps are resolved by minimum Haversine distance. This one-time spatial join materialises the per-site counts $C_{i,s}$ (Eq. (1)) as a static lookup table. The Streamlit Dashboard polls `/api/v1/safety/index` every five minutes and recolors each `folium.CircleMarker`—green (SI $<$ 60), amber (60–75), red ($>$ 75)—with radius proportional to traffic volume, giving operators a dual spatial encoding of risk and exposure (Figure 2a).

III. SafetyChat: LLM-Powered Explanation Module

VTTSI already answers *what* the score is; SafetyChat answers *why* and *what to do*. GPT-4o is equipped with typed function calls [7] that map to VTTSI REST endpoints, grounding every claim in live data. An agentic loop (up to six iterations) enables chained calls—e.g., ranking all intersections then drilling into the top site's breakdown. The six tools are: (1) `get_safety_score` — RT–SI, MCDM, and blended scores with uplift factors; optional `target_time` for historical queries; (2) `get_component_breakdown` — CRITIC-weighted criteria for a 15-min bin; (3) `get_historical_baseline` — EB prior and 7-year crash severity breakdown; (4) `compare_intersections` — ranks all sites by any index component; (5) `get_trend_data` — 7–30 day MCDM trend with sampled snapshots; (6) `run_sql_query` —

TABLE I
VTTSI-Chat vs. Related Systems

System	Hybrid Index	Real-Time Telemetry	LLM Interface	Causal Narrative
Google / Apple Maps	✗	✓	✗	✗
Waycare / ATSPM	✗	✓	✗	✗
TrafficGPT [11]	✗	✓	✓	✗
LLMLight [4]	✗	✓	✓	✗
GPT-Driver [5]	✗	✓	✓	✗
VTTSI-Chat (ours)	✓	✓	✓	✓

TABLE II
Summary of Demonstration Use Cases

Use Case	Key Tool Calls	User
Morning briefing	compare + breakdown	TMC Operator
Causal explanation	score(t) + breakdown(t)	Planner
Responder routing	score $\times$ 2	Dispatcher
Trend engagement	get_trend_data(30d)	Official/Public

read-only ad-hoc SQL, write-guarded.

Table I positions VTTSI-Chat against related systems.

IV. Demonstration

The live demo runs against the production VTTSI deployment, monitoring 18 intersections in Falls Church, VA. Four use cases are demonstrated.

A. UC1: TMC Operator Morning Briefing

Query: "Give me a morning safety briefing for all intersections." SafetyChat chains `compare_intersections` and `get_component_breakdown` for the two highest-ranked sites, delivering a situational-awareness narrative identifying peak-risk windows and dominant criteria (e.g., speed variance, VRU counts) in under five seconds.

B. UC2: Causal Safety Explanation for Planners

Query: "Why did E. Broad–N. Washington score above 70 yesterday at 5 PM?" SafetyChat queries `get_safety_score` and `get_component_breakdown` at the historical time-bin, identifies criteria in the 90th percentile, and traces the elevated score to concurrent peak vehicle volume and incident activity.

C. UC3: Emergency Responder Routing

Query: "Which of Glebe–Potomac or Birch–W. Broad has lower current risk?" SafetyChat compares live RT–SI, MCDM, and blended indices for both intersections and recommends the lower-risk route with reasoning grounded in VRU counts and speed variance.

D. UC4: Public Stakeholder Trend Engagement

Query: "Is the Glebe–Potomac intersection getting safer over time?" SafetyChat calls `get_trend_data` (30-day window) and translates mean/max/min MCDM scores into plain-language risk levels with a trend direction indicator.

V. Related Work

VTTSI's scoring builds on EB-stabilized crash models [8, 6], MCDM-based indices [1, 2], and surrogate safety indicators [10, 3]. In the LLM domain, TrafficGPT [11], LLMLight [4], and GPT-Driver/DriveVLM [5, 12] address flow analysis, signal control, and AV planning respectively—none provide a safety-index explanation interface for infrastructure operators.

VI. Conclusion

VTTSI-Chat closes the gap between quantitative safety scoring and actionable situational awareness by grounding all LLM outputs in six typed API tool calls, delivering causally grounded safety narratives to TMC operators, planners, and emergency responders through natural language dialogue. The spatially-joined crash baseline, bounding-box intersection registry, and live 15-minute telemetry together form the *foundational geospatial layer* for future work integrating the Agentic Traffic Safety Knowledge Graph (TS–KG)—enabling the LLM to reason over intersection geometry, conflict topology, and causal trajectory motifs.

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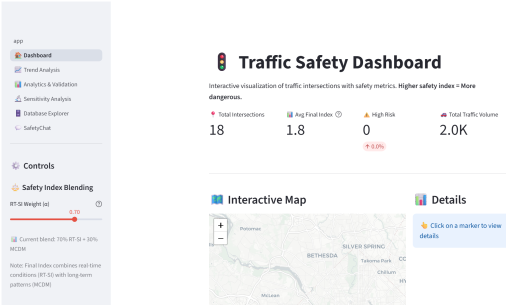
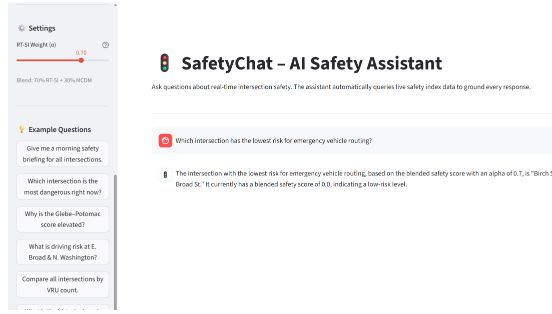


(a) Dashboard: interactive map with 18 monitored intersections (Falls Church, VA), real-time KPI cards (Avg Index 1.8, 0 High-Risk sites, 2.0 K traffic volume).



(b) SafetyChat (UC3): emergency routing query—GPT-4o calls `compare_intersections` *and returns Birch & W. Broad as the safest crossing (blended score 0.0) for immediate vehicle dispatch.*

Fig. 2. Live VTTSI-Chat demonstration: (a) geospatial dashboard providing system-wide situational awareness; (b) SafetyChat natural-language response grounding an emergency routing recommendation in live blended safety indices.

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